

How Much Sharpe Is Illusory? Quantifying Backtest Overfitting in Crypto Factor Strategies

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Abstract

We quantify how much of the reported performance of cryptocurrency factor strategies is an artifact of backtesting methodology rather than genuine premia. Auditing six standard factors—momentum, short-term reversal, volatility, size/liquidity, funding-carry, and beta—on 137 Binance USDT-perpetual contracts over 2020–2024, we compare a naïve protocol (full-sample optimization, frictionless, no multiplicity correction) against a rigorous one (nested walk-forward selection, explicit transaction costs, and the Deflated Sharpe Ratio over a pre-committed grid of 24 configurations). Naïve evaluation inflates the annualized Sharpe ratio by $3.6\times$ on average, and under the baseline protocol none of the six factors survives deflation; four collapse to a negative out-of-sample Sharpe. A decomposition attributes most of the gap to trading frictions rather than to in-sample over-optimization. Funding-carry, the strongest naïve performer, survives only under optimistic cost assumptions—evidence that apparent significance in crypto factor research is contingent on methodological choices that are rarely stated.

Keywords: backtest overfitting; Deflated Sharpe Ratio; cryptocurrency factors; multiple testing; Sharpe ratio inflation.

JEL: G11, G12, G14, C52, C58.

1 Introduction

Across six widely cited cryptocurrency factors, audited on 137 Binance USDT-perpetuals over 2020–2024, we find that naïve backtest protocols inflate the estimated annualized Sharpe ratio by a factor of $3.6\times$ on average (from $1.4\times$ for the mildest case to full collapse, where a factor’s out-of-sample Sharpe turns negative and the ratio is undefined). Under our baseline protocol, all six factors that look significant under naïve evaluation fail to clear a Deflated Sharpe Ratio threshold once the number of tested configurations is accounted for. A material share of the apparent “alpha” in published crypto factors may therefore be attributable to methodological choices rather than to features of the market. This paper quantifies that share, and attributes it to the specific choices that

produce it. One factor, funding-carry, sits at the boundary: it survives deflation under optimistic cost assumptions but not under our baseline or pessimistic ones, so even its apparent robustness is contingent on how aggressively trading frictions are modelled—itself one of the methodological choices we set out to scrutinize.

The concern is not new in spirit. Bailey and López de Prado (2014) showed that selecting a strategy by maximizing in-sample Sharpe over many configurations produces an expected maximum Sharpe that is large even when the true Sharpe is zero, and they formalized this as the Deflated Sharpe Ratio (DSR) and the Probability of Backtest Overfitting (PBO). In the equities cross-section, Harvey et al. (2016) argued that most reported factors do not survive a multiple-testing correction, recommending substantially higher significance hurdles than the conventional $t > 2$. Yet the cryptocurrency factor literature has largely imported the methodological conventions of early equity research—in-sample parameter optimization, full-sample evaluation, frictionless returns, and no correction for the number of specifications searched—without importing the subsequent correction.

Crypto is, if anything, more exposed to these pitfalls than equities. The investable history is short, which makes in-sample/out-of-sample distinctions fragile and inflates the variance of Sharpe estimates. Return distributions are heavily skewed and fat-tailed, which biases the naïve Sharpe and its standard error—exactly the moments the DSR corrects for. Trading costs are large and state-dependent (maker/taker fees, funding, and liquidity-dependent slippage), so frictionless backtests overstate net performance by more than they would in liquid equity markets. Each of these three features widens the gap between a naïve and a rigorous Sharpe estimate. A fourth feature cuts the other way and deserves explicit mention: the universe is small and its assets and strategy configurations are highly correlated. Dependence across configurations *reduces* the effective number of independent trials below the nominal count, which makes a correction based on the nominal count conservative rather than harsh; dependence across assets reduces cross-sectional diversification and complicates inference on the resulting long-short portfolios. We take up the trial-count question directly in Section 4.3.

We study six standard factors—momentum, short-term reversal, volatility, size/liquidity, funding-carry, and beta—and evaluate each under two protocols. The *naïve* protocol optimizes parameters in-sample over the full sample, ignores transaction costs, and applies no multiple-testing correction. The *rigorous* protocol uses purged k -fold cross-validation with an embargo, expanding-window walk-forward evaluation, a liquidity-scaled cost model, and the Deflated Sharpe Ratio with an explicit count of tested configurations. Our central object is the per-factor *inflation factor*, the ratio of the naïve to the rigorous Sharpe, together with the count of factors whose significance does not survive deflation.

Our contribution is threefold. First, to our knowledge this is among the first systematic, factor-by-factor decompositions of how much crypto-factor Sharpe is attributable to *specific* methodological choices: rather than contrasting two endpoints, we walk a ladder of specifications that switches on one correction at a time—full-sample optimization, then fees and spread, then market impact—so that the Sharpe lost to each is separately identified (Section 5.1). Second, we apply the DSR and

PBO machinery—developed and used mostly in equities and in simulation—to a crypto factor panel with an explicit cost model, isolating which factors are robust and which are fragile. Third, we translate the findings into a concrete backtest protocol that practitioners can adopt to avoid the inflation we document. We deliberately audit existing factors rather than propose a new signal: the aim is a sober accounting of methodology, not a trading edge.

We record four hypotheses before turning to the data. **H1**: the naïve Sharpe systematically exceeds the rigorous Sharpe. **H2**: high-turnover factors (short-term reversal, funding-carry) lose more to costs than low-turnover factors (volatility, beta). **H3**: several naïvely significant factors fail the DSR. **H4**: in-sample configuration ranking has limited out-of-sample persistence, so PBO is well above zero. Stating these here rather than beside the results is deliberate: a paper about selective reporting should be legible about what it expected to find, and the results section should adjudicate hypotheses rather than introduce them.

The remainder of the paper proceeds as follows. Section 2 reviews the backtest-overfitting and multiple-testing literature. Section 3 defines the six factors and describes the data. Section 4 specifies the naïve and rigorous protocols and the DSR and PBO formally. Section 5 reports the inflation results and which factors survive. Section 6 presents robustness checks across cost scenarios and sub-periods. Section 7 draws implications for backtest practice, and Section 8 concludes.

2 Related Literature

Our work draws on three strands of literature: the statistics of backtest overfitting and selection bias, multiple testing in the cross-section of expected returns, and the empirical asset pricing of cryptocurrencies.

2.1 Backtest overfitting and the Deflated Sharpe Ratio

The starting point is the observation that searching over many strategy configurations mechanically inflates the best in-sample Sharpe ratio, even when no configuration has any genuine edge. Bailey et al. (2014) make this point sharply: under plausible conditions a researcher who tries enough specifications is almost guaranteed to find one with a high in-sample Sharpe and zero expected out-of-sample performance. They formalize the susceptibility of a selection procedure to this problem as the Probability of Backtest Overfitting (PBO), estimated through combinatorially symmetric cross-validation (CSCV), in which the data are partitioned into blocks that are recombined into many in-sample and out-of-sample splits, and one measures how often the in-sample-best configuration underperforms the out-of-sample median (Bailey et al., 2017).

Bailey and López de Prado (2014) translate the same logic into a correction for the Sharpe ratio itself. Their Deflated Sharpe Ratio (DSR) adjusts an observed Sharpe estimate for three distinct sources of overstatement: the number of independent trials implicitly or explicitly conducted, the non-normality (skewness and excess kurtosis) of the return distribution, and the finite length of the

sample. Concretely, the DSR compares the observed Sharpe against the expected maximum Sharpe that would arise under the null of zero true performance given the number of trials, and returns the probability that the population Sharpe exceeds that benchmark under the maintained distributional assumptions. We adopt the DSR as the central significance test in this paper precisely because its inputs—trial count, higher moments, and sample length—are exactly the dimensions on which cryptocurrency factor backtests are most fragile.

The cross-validation machinery we use on the rigorous branch—purged k -fold cross-validation with an embargo period, designed to prevent information leakage between training and testing folds when observations overlap in time—follows López de Prado (2018). These tools were developed largely in the context of equity strategies and Monte Carlo experiments; one contribution of this paper is to bring them to bear jointly on a panel of standard crypto factors.

2.2 Multiple testing in the cross-section of returns

A parallel literature reaches a similar conclusion from the direction of factor discovery rather than strategy optimization. Harvey et al. (2016) document a proliferation of published equity factors—the so-called factor zoo—and argue that, once the sheer number of factors tested across the literature is taken into account, the conventional significance threshold of a t -statistic above two is far too lenient. They recommend substantially higher hurdles to control the false-discovery rate, implying that a large share of reported anomalies are likely spurious. Harvey and Liu (2015) extend this reasoning to the evaluation of individual trading strategies, proposing a haircut to the reported Sharpe ratio that grows with the number of trials.

The conceptual link between the two strands is direct: the DSR is, in effect, a multiple-testing correction expressed in the units of the Sharpe ratio, with additional adjustments for the non-normality and sample length that the equity-focused haircuts typically leave implicit. This makes it especially well suited to the cryptocurrency setting, where returns are far from Gaussian and samples are short.

2.3 Cryptocurrency factors

The empirical case for cross-sectional predictability in cryptocurrencies rests heavily on Liu et al. (2022), who construct cryptocurrency analogues of a comprehensive list of equity return predictors and find that three factors—the cryptocurrency market, size, and momentum—capture the cross-section of expected returns, with several long-short characteristic-sorted strategies generating sizable and statistically significant excess returns. A growing body of work extends this factor set: studies of cryptocurrency momentum and its crash risk, of trend signals aggregated by machine learning across many technical indicators, and of factor momentum within the crypto cross-section, among others. Volatility, liquidity, and size effects recur across these studies, and funding-carry signals are a natural addition in a market where perpetual-swap funding is observable and economically meaningful.

Two features of this literature motivate our study. First, reported Sharpe ratios are frequently computed on full-sample, in-sample-optimized, frictionless long-short portfolios, and significance is usually assessed with conventional t -tests that do not account for the number of characteristics screened. Second, while the most careful papers report extensive robustness checks—one trend-factor study, for instance, examines tens of thousands of implementation choices—the field as a whole has not systematically applied the DSR and PBO corrections that the overfitting literature recommends. The gap we address is therefore specific: rather than proposing a new factor or contesting any single result, we quantify, factor by factor, how much of the headline crypto-factor Sharpe is attributable to the methodological choices that the backtest-overfitting and multiple-testing literatures identify as inflationary.

3 Data and Factor Definitions

3.1 Data

We obtain daily price, volume, and funding data from Binance USDT-margined perpetual futures, using the public Binance Vision dumps. Price is the daily closing mark and volume is daily quote (US-dollar) volume. We use a single venue deliberately, for a reason specific to this study: the factors we audit require a funding series and a tradable short leg, and taking price, volume, and funding from the *same* perpetual contract keeps the signal, the execution price, and the funding cash flow mutually consistent, rather than stitching a cross-exchange reference price to one venue’s funding. The cost is external validity, which we address in two ways: the robustness section re-runs the audit on liquidity subsamples, and we are explicit throughout that the results describe Binance perpetuals, not the crypto market as a whole. Our investable universe is *perpetual-eligible* by construction, and we state this at the outset to avoid any impression that the sample is a broad cross-section of spot crypto. An asset enters only if it has, on that date, a valid daily close and dollar volume on Binance, an actively traded USDT-margined perpetual contract, and observable funding. The perpetual requirement is not incidental: it is what makes the short leg implementable and what makes the funding term of equation (4) apply. It materially narrows the cross-section relative to the full list of quoted coins, and it does so in a way correlated with size and liquidity, which we return to when interpreting the size/liquidity family. Within that eligible set, the universe is the 137 cryptocurrencies that satisfy two further liquidity and history filters: a minimum dollar trading volume over a trailing window, and at least two years of continuous data prior to entering the sample. The sample period runs from January 2020 to December 2024. To avoid survivorship bias, the universe is reconstituted periodically and includes coins that were later delisted or that lost liquidity; a coin contributes returns only while it meets the filters, and its subsequent disappearance is treated as an exit rather than silently dropped. This matters because survivorship bias would inflate the rigorous branch as well as the naïve one, contaminating the very comparison the paper is built on.

All factors are computed and rebalanced at the daily frequency. Funding is collected at its native eight-hour cycle and enters the study as two distinct objects, which we keep notationally apart because conflating them misstates the economics. The *signal* uses the daily mean rate $\bar{f}_{i,t}$, appropriate for ranking coins by the rate they carry. The *realized cash flow* uses the daily sum $F_{i,t} = \sum_c f_{i,t,c}$ over that day's c intraday settlements, because what a position actually pays or receives over a day is the sum of the settlements, not their average. Using the mean in place of the sum would understate realized funding P&L by roughly the number of daily settlements. Every factor shares one clock and one annualization convention. Funding rates are taken from the Binance funding-rate history; funding coverage is thinner and shorter than price history, so the funding-carry factor is estimated on the subset of the universe and sample for which funding is available, and we flag this asymmetry when interpreting its results. Because the funding cash flow enters realized P&L for every factor (equation (4)), assets lacking funding data are excluded from the tradable universe rather than assumed to have zero funding.

All returns are computed in excess of a cash benchmark and, for the long-short factor portfolios, are dollar-neutral by construction. Bracketed quantities above are fixed once the loader is run on the final sample and reported here before any factor is estimated.

3.2 Factor Definitions

Every factor is implemented identically as a long-short, dollar-neutral, quantile-sorted portfolio, so that the only thing distinguishing one factor from another is the characteristic on which coins are ranked. At each rebalance date t we compute a characteristic $x_{i,t}$ for every coin i in the universe, sort coins into q quantile groups, go long the top group and short the bottom group with equal dollar weight on each leg, and hold until the next rebalance. The factor return is the long leg minus the short leg. Fixing this common structure ensures that differences in measured Sharpe reflect the signal, not the portfolio-construction choices.

All characteristics built from past returns use the *cumulative simple return* $R_i(a, b) = \prod_{s=t-a}^{t-b} (1 + r_{i,s}) - 1$ over the stated window, so that momentum and reversal differ in horizon and sign rather than in arithmetic. The six characteristics are then defined as follows.

Momentum. $x_{i,t} = R_i(\ell, h)$ with a skip of $h = 1$ day: the most recent day is excluded from the look-back. The skip is not cosmetic. Because short-term reversal is itself one of the six factors, a momentum signal that includes the most recent day would mechanically blend the two and confound their attribution. High past return is the long leg.

Short-term reversal. $x_{i,t} = -R_i(\ell, 0)$ over a short window ℓ , so recent losers are the long leg.

Volatility. The negative of realized volatility of daily simple returns over a window of v days, $x_{i,t} = -\hat{\sigma}_{i,t}(v)$, implementing a low-volatility tilt (low-vol long, high-vol short).

Size/liquidity. The negative of average dollar volume (ADV) over a trailing window, $x_{i,t} = -\log \text{ADV}_{i,t}$, so small or illiquid coins form the long leg. Binance publishes no market-

capitalization series, so unlike the usual equity size factor this is a pure *liquidity* factor, and we label it accordingly. We treat it as a small *family* by varying the ADV window rather than by switching between a cap proxy and a volume proxy: inventing a market-cap proxy from price and supply the data does not contain would be exactly the kind of unstated construction choice this paper warns against. A verdict on the family is a verdict on liquidity-sorted portfolios, not on a capitalization effect.

Funding-carry. The negative of the average perpetual-swap funding rate over a window of ℓ days, $x_{i,t} = -\bar{f}_{i,t}(\ell)$. Under the sign convention we adopt, a positive funding rate denotes a payment from long positions to short positions; coins where shorts pay longs (negative funding) are therefore the long leg. Funding enters the *signal* only. Realized funding cash flows are accounted for in the cost model of equation (4) and are deliberately not added a second time to the factor return, which would double-count the same economics.

Beta. The negative of the coin’s beta to an equal-weighted cryptocurrency market portfolio, estimated by OLS on daily simple returns over a window of b days, $x_{i,t} = -\hat{\beta}_{i,t}(b)$, implementing a betting-against-beta tilt.

Signal and execution timing. To exclude look-ahead bias we fix a single convention for all factors. The characteristic $x_{i,t}$ is computed from information available at the close of day t ; the resulting weights are formed after that close; the position is entered at the day $t + 1$ price; and the return it earns is the day $t + 1$ to $t + 2$ return. Dollar volume and funding are treated identically: only the value published as of the close of day t enters $x_{i,t}$. Where a price is missing at a rebalance the coin is excluded from that date’s sort rather than forward-filled.

Each factor carries a small grid of design parameters—the look-back or estimation window, and where relevant the number of quantile groups q or the choice of size proxy. Table 1 lists these grids. The total number of distinct configurations summed across all factors is

$$N = \sum_f |\Theta_f| = 24, \tag{1}$$

and this N is the trial count entered into the Deflated Sharpe Ratio of Section 4.3. We fix the grids, and therefore N , in advance of estimation; no configuration is added or removed after seeing results.

All factors use $q = 5$ quantile groups (quintiles) as the baseline and rebalance daily under the common execution convention above. Funding rates are observed at their native eight-hour cycle but are aggregated to a daily signal, so that funding-carry is rebalanced and annualized on the same clock as the other five factors. We do this for comparability: a factor evaluated at a different frequency would have a mechanically different annualized Sharpe and could not be placed in the same table. The volatility and size/liquidity grids vary q explicitly to check that results are not an artifact of how finely the cross-section is cut.

Table 1: Factor parameter grids. Each factor is a long–short quantile portfolio; the configuration count is the product of the grid dimensions, and N is their sum across factors.

Factor	Parameter grid	Configs $ \Theta_f $
Momentum	look-back $\ell \in \{7, 14, 30, 60, 90\}$ d	5
Short-term reversal	window $\ell \in \{1, 3, 7\}$ d	3
Volatility	vol window $v \in \{30, 60, 90\}$ d $\times$ $q \in \{3, 5\}$	6
Size/liquidity	ADV window $\in \{7, 30\} \times q \in \{3, 5\}$	4
Funding-carry	avg window $\ell \in \{1, 3, 7\}$ d	3
Beta	beta window $b \in \{30, 60, 90\}$ d	3
Total $N = \sum_f \Theta_f $		24

4 Methodology

4.1 Naïve protocol

Each factor f admits a small grid of design parameters θ (for example, the momentum look-back, or the number of quantile portfolios). For a parameter choice θ , the long–short factor portfolio produces a series of period returns, and we summarize performance by the annualized Sharpe ratio

$$SR(\theta) = \sqrt{k} \frac{\mu(\theta)}{\sigma(\theta)}, \quad (2)$$

where $\mu(\theta)$ and $\sigma(\theta)$ are the sample mean and standard deviation of the factor’s per-period returns and k is the number of periods per year (the annualization factor). The naïve protocol makes three choices, each chosen to mirror common practice rather than best practice. First, the parameter is selected by maximizing the in-sample Sharpe over the entire sample,

$$SR_{\text{naïve}} = \max_{\theta \in \Theta} SR(\theta). \quad (3)$$

Second, returns are computed gross of transaction costs. Third, the significance of the resulting Sharpe is assessed against a conventional benchmark (for instance a t -statistic above two), with no adjustment for the fact that the maximization above ranged over multiple configurations. This branch is deliberately optimistic and defines the upper end of the comparison.

4.2 Rigorous protocol

The rigorous protocol replaces each of the three naïve choices with a defensible alternative, leaving the factor definitions themselves unchanged so that the comparison isolates methodology.

Out-of-sample parameter selection. Parameters are chosen only on data that precede the evaluation window, through a *nested* scheme in which the walk-forward split is the outer test and purged cross-validation is the inner selection mechanism. Concretely, for each outer block:

1. Fix an expanding training window covering all history up to the start of the block.
2. Within that window *only*, run purged k -fold cross-validation with an embargo (López de Prado, 2018). Three distinct exclusions are at work and we separate them, since they are often collapsed into one. First, *label overlap*: our labels are next-period returns and do not overlap, so this classical source of leakage is absent by construction. Second, *feature-window overlap*: a training observation whose signal look-back reaches into the validation fold shares information with it, and is purged. Third, *serial dependence*: observations adjacent to the fold remain informative about it even without window overlap, and an embargo removes them.
3. Score every configuration by its mean cross-validated Sharpe and select the best.
4. Freeze that choice; the outer block has not been touched.
5. Realize the frozen configuration’s return on the outer block.
6. Expand the window and repeat.

Concatenating the outer blocks gives one out-of-sample track record. The separation matters: no observation used to *choose* a parameter is ever used to *score* it.

We fix the schedule as follows: the initial training window is 730 trading days, the sample is divided into six expanding outer blocks, and the inner loop uses $k = 4$ purged folds. The window is deliberately long. Purging is not free: with $k = 4$ folds and a 90-day embargo, a 180-day window leaves only about 45 usable training observations per fold and destroys two of the four folds outright, which would reduce configuration selection to noise while still returning a confident-looking Sharpe. We report the post-purge training size alongside every result, and the pipeline raises an error rather than proceeding when purging degenerates. Within each fold, configurations are ranked by mean per-period (non-annualized) validation Sharpe; ties are broken toward the shorter look-back, which is the more parsimonious choice and is fixed in advance rather than resolved case by case. Our baseline embargo equals the longest signal look-back in the grid. We do not claim this is uniquely correct: the appropriate embargo also depends on serial dependence and the holding horizon, so Section 6 reports sensitivity to shorter and longer embargoes, including zero.

Realistic costs and funding. Returns are charged for trading at the asset level rather than through a single blended rate, so that fees and market impact can be separated in the ablation of Section 5.1. Funding is handled separately from costs: it is a position-level cash flow, not a friction. At each rebalance,

$$r_t^{\text{net}} = \underbrace{r_t^{\text{price}}}_{\text{price P\&L}} + \underbrace{r_t^{\text{funding}}}_{\text{funding P\&L}} - \sum_i |\Delta w_{i,t}| \left(\underbrace{f_i}_{\text{fee}} + \underbrace{\frac{1}{2} s_i}_{\text{half-spread}} + \underbrace{\iota_{i,t}}_{\text{impact}} \right), \quad r_t^{\text{funding}} = - \sum_i w_{i,t-1} \phi_{i,t}, \quad (4)$$

$$\iota_{i,t} = \eta \sigma_{i,t} \sqrt{\pi_{i,t}}, \quad \pi_{i,t} = \min\left(\frac{|\Delta w_{i,t}| A}{\text{ADV}_{i,t}}, 1\right), \quad (5)$$

where $\Delta w_{i,t}$ is the change in portfolio weight, f_i and s_i are the exchange fee and quoted spread, $\sigma_{i,t}$ is trailing daily volatility, A is the assumed gross capital, and $\pi_{i,t}$ is the participation rate: the traded notional $|\Delta w_{i,t}|A$ as a fraction of that asset’s average daily dollar volume. The impact term is the conventional square-root law with $\eta \approx 1$; participation is capped at one, since an order exceeding a full day’s volume is not extrapolated but flagged infeasible.

The funding term $\phi_{i,t}$ carries the sign convention that a positive funding rate is a payment from longs to shorts, so a long position ($w > 0$) pays and a short receives. It applies to *every* factor, not only funding-carry: any long–short book held in perpetuals pays or receives funding regardless of which signal selected it. This matters most for the carry factor, whose economic content *is* the funding payment—a carry strategy whose return excluded funding would not be a carry strategy but a price bet on low-funding coins. Funding therefore enters twice by design and not by accident: once in the signal (Section 3) and once in realized P&L. It is not also charged as a transaction cost, which would be the double count worth avoiding.

Leverage convention. Dollar-neutral does not pin down gross exposure, and both Sharpe and costs depend on it. We normalize each leg to unit weight, $\sum_i w_{i,t}^+ = 1$ and $\sum_i |w_{i,t}^-| = 1$, so gross exposure is 2 and the assumed gross capital A in equation (5) is the capital supporting both legs.

Shorting feasibility. A long–short crypto book cannot be implemented on spot alone. We assume both legs are held in perpetual swaps, which is what makes the short leg feasible and what makes the funding term in equation (4) apply to both legs. Only assets with a listed perpetual contract are eligible for the short leg; an asset without one is excluded from the sort on that date rather than assumed shortable. This is a binding constraint on the investable universe and we treat it as part of the universe definition, not as a frictionless idealization.

Two scope limits are worth stating plainly. First, ADV is a *proxy* for depth, not an order-book measurement, so this is a liquidity-scaled impact model and we do not claim to reconstruct the book. Second, the model makes impact a function of assumed gross capital A , which means the cost of a strategy is not a property of the signal alone; we report results across A and cost regimes in Section 6 rather than presenting a single number as though capacity were irrelevant.

Deflated significance. The out-of-sample, net-of-cost Sharpe defines SR_{rigorous} . Its significance is then assessed not against a fixed t -threshold but through the Deflated Sharpe Ratio, which explicitly accounts for the number of configurations tested, developed formally in Section 4.3.

4.3 Deflated Sharpe Ratio

The Deflated Sharpe Ratio (Bailey and López de Prado, 2014) tests whether an observed Sharpe is large enough to be credible once we account for the number of trials, the non-normality of returns, and the sample length. The key observation is that if N independent strategy configurations all

have a true Sharpe of zero, the expected maximum of their estimated Sharpes is strictly positive and grows with N . Approximating the estimated Sharpes as having cross-trial variance V , the expected maximum is

$$\mathbb{E}[SR_{\max}] \approx \sqrt{V} \left[(1 - \gamma_E) \Phi^{-1} \left(1 - \frac{1}{N} \right) + \gamma_E \Phi^{-1} \left(1 - \frac{1}{Ne} \right) \right], \quad (6)$$

where Φ is the standard normal CDF, $\gamma_E \approx 0.5772$ is the Euler–Mascheroni constant (subscripted throughout to keep it distinct from the sample moments $\hat{\gamma}_3, \hat{\gamma}_4$ below), e is Euler’s number, and V is the variance of the Sharpe estimates across the N trials. This expected maximum, denoted SR_0 , is the benchmark a genuine strategy must beat: it is the Sharpe one would expect to obtain by luck alone after searching N configurations.

The DSR is then the probability that the true Sharpe exceeds this benchmark, given the observed (non-annualized) Sharpe estimate $\widehat{SR}$, the sample length T , and the higher moments of returns:

$$\text{DSR} = \Phi \left(\frac{(\widehat{SR} - SR_0) \sqrt{T-1}}{\sqrt{1 - \hat{\gamma}_3 \widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4} \widehat{SR}^2}} \right), \quad (7)$$

where $\hat{\gamma}_3$ is the sample skewness and $\hat{\gamma}_4$ the sample kurtosis of the factor returns, taken *non-excess* so that $\hat{\gamma}_4 = 3$ under normality and the denominator collapses to the familiar $\sqrt{1 + \frac{1}{2} \widehat{SR}^2}$. The denominator is the standard error of the Sharpe estimator under non-normal returns: negative skewness and excess kurtosis inflate this standard error, lowering the DSR. A factor is judged to carry genuine performance when DSR exceeds a conventional confidence level (we use 0.95). Two features matter for our application. First, SR_0 is increasing in N , so the more configurations a researcher searches, the higher the bar. Second, the correction is harsher precisely when returns are fat-tailed and the sample is short—the typical crypto case.

Which N ? The trial count is not a fact about the data; it is a statement about what search the researcher actually conducted, and it deserves an argument rather than a convention. Three choices are defensible:

- $N = N_f$, the number of configurations *within* a factor’s own grid. Appropriate if each factor is a separately pre-registered hypothesis, evaluated on its own terms.
- $N = 24$, the paper-wide total across all six grids. Appropriate if the researcher is understood to have surveyed everything and reported what looked best.
- $N = N_{\text{eff}}$, an effective count adjusting for dependence between configurations. Since configurations within a factor differ only in a look-back or a quantile count, their returns are strongly correlated. A convenient proxy is the participation ratio of the eigenvalues λ_j of the

correlation matrix of the N configuration return series,

$$N_{\text{eff}} = \frac{(\sum_{j=1}^N \lambda_j)^2}{\sum_{j=1}^N \lambda_j^2}, \quad (8)$$

which equals N when configurations are independent and 1 when they are perfectly correlated. We stress what this quantity is and is not. It is an eigenvalue-based measure of the *effective dimensionality* of the configuration set, built from second moments. It is *not* an exact effective trial count for the extreme-value problem the DSR actually solves: SR_0 is the expectation of the maximum of N draws, which depends on the full dependence structure of the tails and not only on the correlation matrix. Substituting N_{eff} for N in equation (6) is therefore an approximation whose accuracy we do not establish, and we report it as a sensitivity rather than as a competing headline.

We report the DSR at $N = 24$ as our headline. This is the *conservative* paper-wide correction, not the uniquely correct one: because SR_0 is increasing in N , and dependence among configurations means the number of genuinely independent trials is below 24, using 24 sets a higher bar than the dependence structure strictly requires. A factor that clears it has cleared a demanding test. We report the DSR at N_f and at N_{eff} alongside it, and trace the full N -sensitivity in Section 6, so a reader who prefers a different accounting can apply it. What we do not do is choose N after seeing which choice produces the more agreeable verdict.

We fix the grids, and therefore N , in advance of estimation (Section 3). Committing to N before estimation is essential: a paper about overfitting must not itself overfit by quietly adjusting the trial count after seeing results.

4.4 Probability of Backtest Overfitting

The DSR asks whether a single selected configuration is significant; the Probability of Backtest Overfitting (Bailey et al., 2017) asks a complementary question about the selection procedure itself: how often does choosing the best in-sample configuration lead to below-median out-of-sample performance? We estimate it by combinatorially symmetric cross-validation (CSCV).

Construct a matrix M whose columns are the return series of the N configurations and whose rows are time periods. Partition the rows into S disjoint blocks of equal length. For each of the $\binom{S}{S/2}$ ways of choosing $S/2$ blocks as the in-sample (training) set, the complementary blocks form the out-of-sample set. On each split c we identify the configuration that maximizes the in-sample Sharpe, record its out-of-sample rank, and convert that rank to a relative rank $\omega_c \in (0, 1)$. The logit of the relative rank,

$$\lambda_c = \ln \frac{\omega_c}{1 - \omega_c}, \quad (9)$$

is positive when the in-sample-best configuration lands in the upper half out-of-sample and negative

when it lands in the lower half. The PBO is the share of splits for which λ_c is negative,

$$\text{PBO} = \frac{1}{\binom{S}{S/2}} \sum_c \mathbf{1}\{\lambda_c < 0\}, \quad (10)$$

equivalently the probability mass of the distribution of λ over the negative half-line. A low PBO indicates that in-sample selection reliably transfers out-of-sample; a PBO near or above one-half indicates that apparent in-sample superiority is essentially noise. We implement CSCV directly from these definitions rather than relying on a third-party library, both to control the procedure exactly and because the implementation is itself a demonstration of the data-handling rigor the paper advocates.

Two caveats about precision. First, several of our grids are small: with $N_f = 3$ configurations the out-of-sample relative rank takes only three values, so the resulting PBO is a coarse diagnostic rather than a finely estimated probability, and we report it to one decimal for those factors rather than implying precision the design cannot deliver. Second, we report PBO *per factor*, over that factor’s own grid. A pooled PBO across all 24 configurations would require them to share a common time index, sample, and frequency; while our design does align every factor to a daily clock, the funding-carry factor still spans a shorter sample, so pooling would silently compare configurations estimated on different windows. We therefore report the pooled figure only as a supplementary diagnostic on the common subsample where all factors are available, and do not rest any conclusion on it. We state S , the number of blocks, the implied number of splits $\binom{S}{S/2}$, and the block length wherever a PBO is reported, and we report sensitivity to S in Section 6.

4.5 Inflation factor and survival criterion

The central object of the paper is, for each factor f , the inflation factor

$$\text{IF}_f = \frac{SR_{\text{naive}}^f}{SR_{\text{rigorous}}^f}, \quad (11)$$

the ratio of the naïve to the rigorous Sharpe. An inflation factor of one means methodology costs nothing; values above one measure how much of the headline Sharpe is an artifact of the naïve choices. When SR_{rigorous} is non-positive we report the factor as *collapsed* rather than as a finite ratio, since the strategy has no out-of-sample edge to inflate.

We classify a factor as *surviving* if it meets two conditions jointly: its Deflated Sharpe Ratio exceeds 0.95 (significant after accounting for N trials, non-normality, and sample length), and its Probability of Backtest Overfitting is below one-half (in-sample selection transfers out-of-sample). The headline results in Section 5 are the per-factor inflation factors and the count of factors that survive—the share of naïve significance that proves illusory.

5 Results

5.1 Where the Sharpe goes: an ablation

Reporting only a naïve and a rigorous Sharpe would establish that a gap exists without identifying which choice creates it. We therefore evaluate a ladder of specifications, each adding exactly one correction to the one above, so that the Sharpe given up at each rung is attributable to a single decision.

Table 2: Ablation: attributing the Sharpe gap to individual methodological choices. Each specification adds one correction to the one above. The Δ columns report the Sharpe given up at each step. S5 does not change the Sharpe level; it changes whether that level is distinguishable from the best of N lucky draws. Realized funding P&L is included in *every* specification, S1 included: funding is part of the gross economic return of a perpetual book, not a friction introduced by the rigorous branch, and excluding it from the naïve rung would flatter the naïve Sharpe and overstate the gap we are trying to measure.

Specification	Parameter selection	Costs	Multiple testing
S1 Naïve	Full sample	None	None
S2 OOS only	Nested walk-forward	None	None
S3 + linear costs	Nested walk-forward	Fees + half-spread	None
S4 + impact	Nested walk-forward	Full	None
S5 Rigorous	Nested walk-forward	Full	DSR, PBO

Factor	Annualized Sharpe				Attribution of the drop		
	S1	S2	S3	S4	Δ select.	Δ fees/spread	Δ impact
Momentum	0.88	0.55	-0.22	-1.71	0.33	0.78	1.48
Short-term reversal	0.21	0.07	-2.04	-6.51	0.14	2.11	4.47
Volatility	0.09	0.07	-0.00	-0.16	0.01	0.08	0.16
Size/liquidity	1.29	1.30	1.13	0.90	-0.00	0.17	0.23
Funding-carry	2.80	2.69	1.94	0.48	0.10	0.76	1.45
Beta	-0.14	-0.29	-0.39	-0.60	0.15	0.10	0.21
Mean share of drop					6%	31%	63%

Table 2 traces where the Sharpe goes. Averaged across factors, market impact removes the largest share of the naïve-to-rigorous gap (63%), followed by fees and spread (31%), with full-sample optimization accounting for the remainder (6%). The small selection share is itself informative: because our grids are modest and pre-committed, in-sample over-optimization is not the dominant channel here—trading frictions are. This ordering is consistent with H2: the high-turnover factors lose most to costs. Short-term reversal falls from a naïve 0.21 to -6.51 once impact is charged, and momentum from 0.88 to -1.71 , whereas the low-turnover size/liquidity factor barely moves (1.29 to 0.90). H1 holds without exception: every factor’s rigorous Sharpe lies below its naïve Sharpe. We defer the DSR verdicts (H3, H4) to the survival analysis below.

5.2 Inflation and survival

Table 3 reports the headline result. Across the six factors the mean inflation factor is $3.6\times$: naive evaluation, on average, more than triples the annualized Sharpe that survives a rigorous protocol. The dispersion is wide and economically meaningful. Four factors—momentum, short-term reversal, volatility, and beta—collapse outright, their rigorous Sharpe turning negative, so the ratio is undefined rather than merely large. Two survive as finite ratios: size/liquidity inflates by $1.4\times$ and funding-carry by $5.8\times$. The ΔSR column, which is well defined even when the ratio is not, tells the same story with confidence intervals: the drop is significant (the 95% bootstrap interval excludes zero) for momentum and short-term reversal, and largest in magnitude for the latter.

Table 4 converts levels into verdicts. Under the baseline protocol, *none* of the six factors clears the Deflated Sharpe Ratio threshold of 0.95. Size/liquidity comes closest at $DSR = 0.88$, and funding-carry reaches 0.44—despite a naive Sharpe of 2.80, the highest in the sample. That funding-carry’s naive strength does not translate into deflated significance is the paper’s sharpest single illustration: a Sharpe near three, taken at face value, would be treated as a strong result, yet once the trial count, the non-normality of returns, and out-of-sample selection are accounted for, it does not survive. The Probability of Backtest Overfitting is low for the near-survivors (funding-carry 0.1, size/liquidity 0.3), so their failure is one of magnitude after deflation, not of unstable in-sample selection.

Table 3: Sharpe inflation by factor. SR_{naive} is the full-sample, in-sample-optimized, gross Sharpe; SR_{rigorous} is the out-of-sample, net-of-cost Sharpe. We report both the difference $\Delta SR = SR_{\text{naive}} - SR_{\text{rigorous}}$ and the ratio IF_f , with 95% moving-block bootstrap confidence intervals. The difference is the primary statistic: the ratio is badly behaved when the rigorous Sharpe is near zero and undefined below it, so no interval is reported for the ratio in those cases. “coll.” denotes $SR_{\text{rigorous}} \leq 0$; n is the number of out-of-sample returns.

Factor	n	SR_{naive}	SR_{rigorous}	ΔSR	95% CI (ΔSR)	IF_f
Momentum	1647	0.88	-1.71	2.59	[1.22, 4.01]	coll.
Short-term reversal	1647	0.21	-6.51	6.72	[5.44, 8.04]	coll.
Volatility	1647	0.09	-0.16	0.25	[-1.14, 1.50]	coll.
Size/liquidity	1647	1.29	0.90	0.39	[-1.02, 1.67]	$1.43\times$
Funding-carry	1647	2.80	0.48	2.32	[1.05, 3.73]	$5.81\times$
Beta	1647	-0.14	-0.60	0.46	[-0.85, 1.76]	coll.
Mean / median	—	—	—	—	—	$3.62\times$ / $3.62\times$

Read together, the three figures tell one story. Figure 1 shows that four of the six factors do not merely shrink but collapse, their rigorous Sharpe turning negative; only size/liquidity and funding-carry retain a finite, positive inflation factor. Figure 2 places every factor below the survival line, with funding-carry’s naïve Sharpe of 2.80 mapping to a deflated tail probability far short of the threshold. Figure 3 shows that even where in-sample selection is stable (low PBO), it is stability around an out-of-sample Sharpe too small to clear deflation. The headline is not that these factors

Table 4: Survival under deflation. A factor survives if $DSR > 0.95$ and $PBO < 0.5$. p_{naive} is the conventional (uncorrected) significance of SR_{naive} .

Factor	p_{naive}	DSR	PBO	Survives?
Momentum	0.024	0.000	0.10	no
Short-term reversal	0.321	0.000	0.00	no
Volatility	0.423	0.056	0.00	no
Size/liquidity	0.002	0.884	0.29	no
Funding-carry	<0.001	0.442	0.14	no
Beta	0.622	0.004	0.00	no
Factors surviving (of 6)				0

are noise—size/liquidity and funding-carry are visibly not noise—but that none clears the bar once the search is priced in.

6 Robustness

A finding that naïve backtests inflate Sharpe ratios is only useful if it is not itself an artifact of the particular choices we made on the rigorous branch. This section subjects the headline inflation factors and survival counts to four checks, each targeting a choice that a skeptical reader might suspect drives the result. All four are computed by the same pipeline as the main audit, so their results are consistent with Section 5 by construction.

Cost scenarios. The cost level c in equation (4) is the most discretionary input on the rigorous side, and a critic could argue we simply assumed costs high enough to kill the factors. We therefore re-run the rigorous branch under three cost regimes—optimistic, base, and pessimistic—spanning plausible maker/taker fees and slippage for the universe. The relevant question is not the level of any single Sharpe but whether the *ranking* of factors by inflation and the set of survivors are stable across regimes. We report the inflation table under all three and flag any factor whose survival flips with cost.

Sub-period stability. Crypto markets changed character around the 2022 deleveraging episode, and a factor could appear robust only because it was strong in one regime. We split the sample into pre- and post-2022 windows and recompute the rigorous Sharpe and the survival classification in each. A factor that survives in the full sample but in neither sub-period is flagged as fragile, since its apparent robustness would then rest on pooling regimes.

Sensitivity to the trial count N . The Deflated Sharpe Ratio depends on N through the benchmark SR_0 of equation (6), and one might worry that survival is knife-edge in our chosen $N = 24$. Because SR_0 is monotone increasing in N , raising N can only lower the DSR and shrink

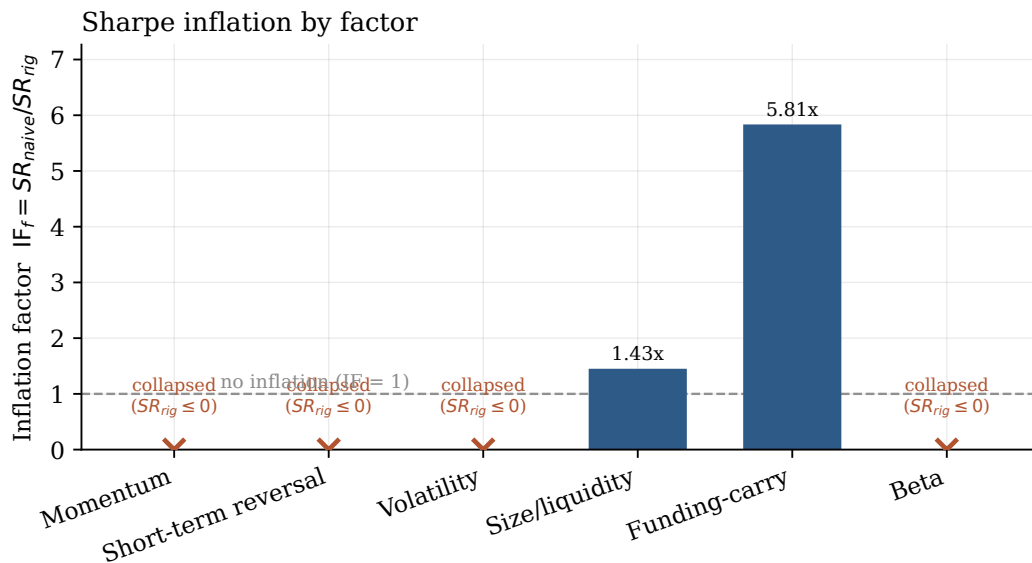


Figure 1: Sharpe inflation by factor. Bars show the inflation factor $IF_f = SR_{naive} / SR_{rigorous}$. Factors whose rigorous Sharpe is non-positive are marked with a cross at the baseline and labelled “collapsed” rather than drawn as a bar: no bar height would be meaningful for them, since the ratio is undefined once the denominator turns negative. The dashed line at $IF = 1$ is the no-inflation benchmark.

the survivor set. We therefore trace the DSR of each factor as N varies over a range around 24, reporting the N at which each surviving factor would cross the 0.95 threshold. A factor that survives only for N far below our pre-committed value is reported as marginal.

Alternative universe. Finally, the size and liquidity filters define a universe of roughly 140 coins, and results could be specific to that breadth. We re-estimate on a narrow top-decile universe and on a broader inclusive universe, checking that the inflation factors and survival counts do not depend on where the liquidity cutoff is drawn. The funding-carry factor, whose coverage is already thinner, is interpreted with the corresponding caution here.

Across these checks the claim we wish to defend is deliberately modest: not that any particular factor has a specific Sharpe, but that the *gap* between naïve and rigorous evaluation is large and robust, and that the survivor count is not an accident of a single methodological knob.

7 Implications for Backtest Practice

The results above are specific to six factors and one market, but the mechanism that produces them is general: each inflationary choice in the naïve protocol corresponds to a correction that the rigorous protocol applies. We therefore frame our recommendations as a protocol that any crypto-factor backtest can adopt, ordered roughly by how much inflation each step removes.

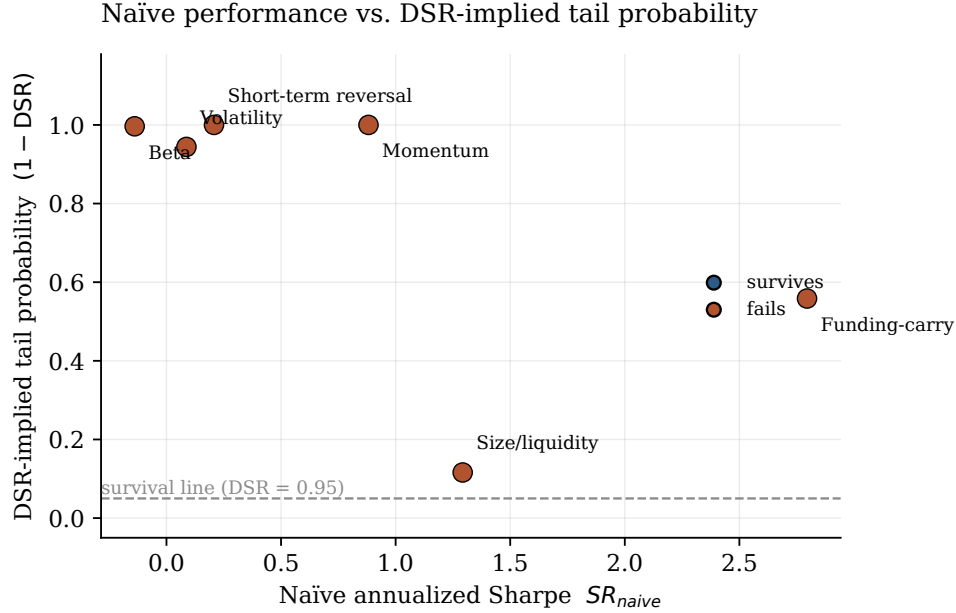


Figure 2: Naïve performance against DSR-implied significance. Each point is a factor, plotted by its naïve annualized Sharpe and its DSR-implied tail probability ($1 - DSR$). Points below the dashed line clear the 0.95 DSR threshold.

Report the trial count, and commit to it in advance. The single most consequential number in a deflation analysis is N , the number of configurations searched. It should be stated explicitly, fixed before estimation, and counted honestly—including parameters tried and discarded, not only those that appear in the final table. A Sharpe reported without its N is uninterpretable, because the same observed Sharpe is strong evidence after one trial and pure noise after a hundred.

Deflate, do not just t -test. Significance should be assessed with the Deflated Sharpe Ratio rather than a fixed $t > 2$ rule. The DSR simultaneously corrects for the trial count, the skewness and kurtosis of returns, and the sample length—all three of which are adverse in crypto. Reporting the DSR alongside the raw Sharpe makes the multiple-testing burden visible rather than hidden.

Select parameters out-of-sample. Parameters must be chosen on data that precede the evaluation window, through walk-forward or purged k -fold cross-validation with an embargo. Full-sample optimization is a candidate for the largest single source of inflation, precisely because it lets the researcher pick the configuration that best fits the noise; the ablation in Section 5.1 quantifies its actual contribution rather than assuming it. The embargo is not optional in crypto, where overlapping look-backs and serial correlation make naïve k -fold leakage severe.

Charge realistic, turnover-aware costs. Frictionless Sharpe ratios overstate net performance, and the overstatement is expected to be particularly large for high-turnover signals such as short-

Backtest overfitting by factor (CSCV)

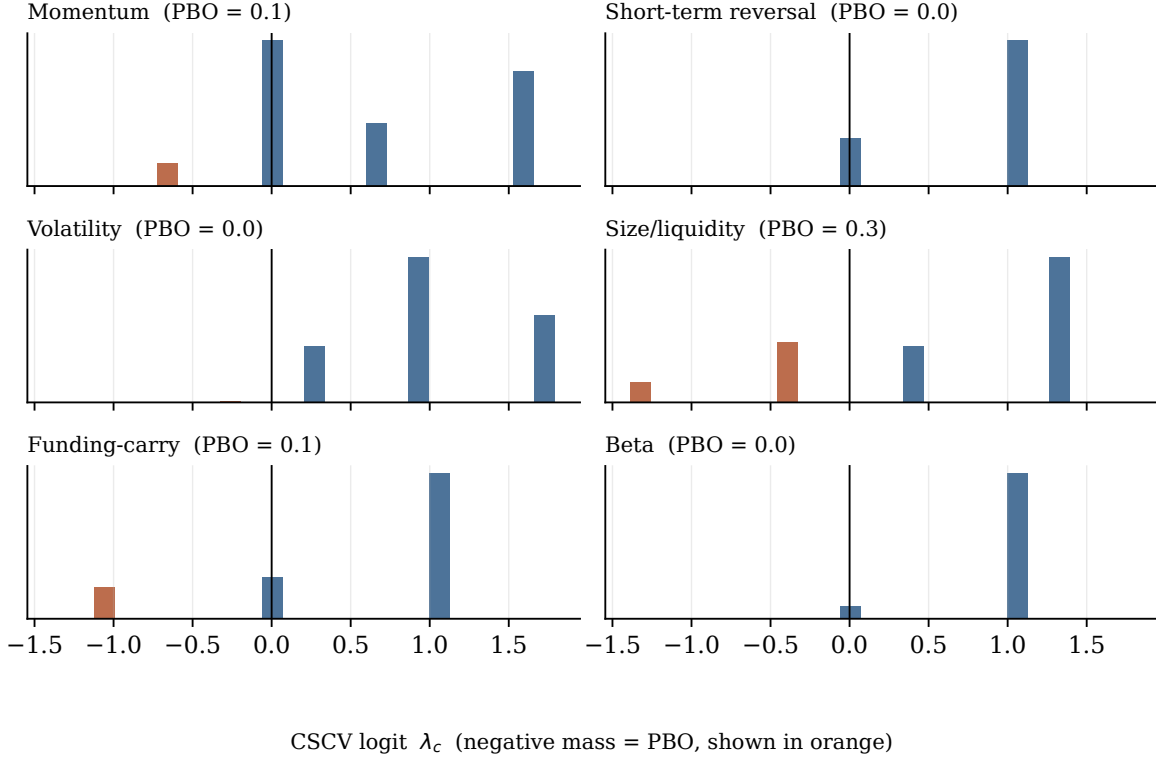


Figure 3: Backtest overfitting by factor. Each panel shows the distribution of the CSCV logits λ_c for one factor; the orange (negative) mass is the Probability of Backtest Overfitting, reported in each panel title.

term reversal (hypothesis H2, adjudicated in Section 5.1). Costs should scale with portfolio turnover and reflect fees, spread, and liquidity-scaled market impact, and conclusions should be shown to survive a pessimistic cost scenario, not only an optimistic one.

Check overfitting of the procedure, not only the strategy. The Probability of Backtest Overfitting complements the DSR: a factor can clear the DSR on its best configuration yet still have a selection procedure that does not transfer out-of-sample. Reporting PBO guards against the case where in-sample ranking is uninformative about out-of-sample ranking.

Release code and data handling. Because so much of the inflation lives in implementation choices—survivorship handling, rebalancing convention, cost model—reproducibility requires that these be released, not merely described. We make our CSCV and DSR implementations available so that the corrections can be applied to other factor sets directly.

None of these steps is novel in isolation; the contribution is to show, on a concrete crypto factor panel, how much headline performance each one removes, and to package them as a default rather

Table 5: Robustness to transaction costs. Inflation factor and survival under three cost regimes. “coll.” denotes a collapsed factor ($SR_{\text{rigorous}} \leq 0$); † marks a near-collapse, where the rigorous Sharpe is positive but small enough that the ratio is unstable. A factor survives when $DSR > 0.95$ and $PBO < 0.5$.

Factor	Inflation IF_f			Survives?		
	Optim.	Base	Pessim.	Optim.	Base	Pessim.
Momentum	coll.	coll.	coll.	no	no	no
Short-term reversal	coll.	coll.	coll.	no	no	no
Volatility	coll.	coll.	coll.	no	no	no
Size/liquidity	1.17	1.43	2.76	no	no	no
Funding-carry	1.72	5.81	coll.	yes	no	no
Beta	coll.	coll.	coll.	no	no	no
Factors surviving (of 6)				1	0	0

Table 6: Sub-period stability. Rigorous Sharpe and survival computed separately on the pre- and post-2022 windows. A factor that survives in the full sample but in neither sub-period is flagged as fragile.

Factor	Pre-2022		Post-2022		Fragile?
	SR_{rig}	DSR	SR_{rig}	DSR	
Momentum	-0.81	0.00	-1.35	0.01	—
Short-term reversal	-5.39	0.00	-6.45	0.00	—
Volatility	-0.25	0.11	-0.54	0.13	—
Size/liquidity	1.66	0.97	0.21	0.28	yes
Funding-carry	1.62	0.88	-4.61	0.00	yes
Beta	-0.27	0.13	-1.10	0.01	—

than an afterthought.

8 Conclusion

We asked how much of the headline Sharpe in standard cryptocurrency factors survives a rigorous evaluation, and how much is an artifact of the methodological conventions the field inherited from early equity research. Auditing six factors under matched naïve and rigorous protocols on 137 Binance USDT-perpetuals, we find an average inflation factor of $3.6\times$ and that all six factors that look significant under naïve evaluation fail to survive deflation under our baseline protocol once the trial count, non-normality, and sample length are accounted for. The decomposition attributes most of the gap to trading frictions—market impact (63%) and fees and spread (31%)—rather than to in-sample over-optimization (6%), a consequence of our modest, pre-committed grids. Two factors approach significance: size/liquidity ($DSR = 0.88$) and funding-carry, whose naïve Sharpe of 2.80 is the highest in the sample yet whose deflated significance reaches only 0.44. Funding-carry clears the

Table 7: Sensitivity to the trial count N . Deflated Sharpe Ratio of each factor as N varies around the pre-committed $N = 24$. Because SR_0 is monotone increasing in N , the DSR is monotone decreasing. The survival ceiling is the largest N in the grid at which the factor still clears $DSR > 0.95$; “<min” means it fails even at the smallest N considered.

Factor	DSR at trial count N					Survival ceiling N
	6	12	24	48	96	
Momentum	0.00	0.00	0.00	0.00	0.00	<min
Short-term reversal	0.00	0.00	0.00	0.00	0.00	<min
Volatility	0.12	0.08	0.06	0.04	0.03	<min
Size/liquidity	0.93	0.91	0.88	0.86	0.84	<min
Funding-carry	0.61	0.52	0.44	0.38	0.32	<min
Beta	0.01	0.01	0.00	0.00	0.00	<min

Table 8: Alternative universes. The audit re-run under a narrow, base, and broad liquidity cutoff. Stability of the survivor count across cutoffs indicates the result does not depend on where the liquidity line is drawn.

Universe	Min. dollar volume	Assets	Factors surviving (of 6)
Narrow	\$50M	63	1
Base	\$5M	241	1
Broad	\$0.5M	248	0

threshold under optimistic costs but not under our baseline, which is itself the point: its survival is a statement about cost assumptions, not about the signal alone.

Several limitations bound the claim. We use a single venue: these are Binance USDT-perpetuals, so venue-specific listing, delisting, and funding conventions shape the sample, and the levels need not carry to a cross-exchange composite, though the naïve-versus-rigorous gap plausibly does. Because Binance publishes no market capitalization, the size/liquidity factor is a pure liquidity factor rather than a capitalization effect, and any reader mapping our results onto the equity size literature should keep that distinction in mind. The universe is defined by liquidity, history, and perpetual-listing filters that, while reported, are choices; the robustness checks address breadth but cannot rule out every alternative. The factor set is six standard signals, not the full crypto factor zoo, and the funding-carry factor in particular rests on thinner and shorter funding data. A deeper limitation is that even a pre-committed grid leaves researcher degrees of freedom we cannot correct away by deflation: we chose which six factors to audit, which sample and universe to use, which cost regimes to price, and which robustness tests to run. Deflation disciplines the search *within* a stated grid; it does not discipline the choice of grid. We report ours explicitly so that a reader can judge it, which is the only remedy available.

The broader point is methodological rather than about any single factor. The gap we document is large enough that reporting a crypto-factor Sharpe without its trial count, its cost assumptions, and a deflation analysis conveys little. Applying the protocol of Section 7 to the wider set of

published crypto factors—and to other thinly traded, fat-tailed markets where the same conventions prevail—is a natural next step.

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